UMAR AHMED THAMEEM AHMED

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EDUCATION

University of Colorado Boulder — **Boulder, CO**
Master of Science – Computer Science — *Aug 2025 – Expected May 2027*

- Coursework: Distributed Systems, Datacenter Scale Computing, Big Data Architecture, Linux System Administration, ML Systems
- CGPA: 4.0/4.0

College of Engineering Guindy, Anna University — **Chennai, India**
Bachelor of Engineering – Computer Science and Engineering — *Aug 2018 – Apr 2022*

- Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Networks, Object-Oriented Programming
- CGPA: 8.75/10 (First Class with Distinction)

WORK EXPERIENCE

F5, Inc. — **Boulder, CO**
Software Development Engineer Intern — *May 2026 – Aug 2026*
Tech Stack: Go, gRPC, Kubernetes, Terraform, AWS

- Implemented cost-efficient autoscaling via AWS ASGs for the NGINX SaaS platform, writing load tests to simulate traffic spikes of **300K concurrent connections** and validate automated scale-out/scale-in triggers
- Achieved **zero dropped connections** during simulated scale-in events by configuring lifecycle hooks and systemd units to manage graceful NGINX shutdown and connection draining
- Accelerated pre-launch performance tuning by building real-time Grafana dashboards with Alloy metrics to track autoscaling resource utilization and projected operational costs

Nueromind Technologies — **Bangalore, India**
Founding Software Engineer — *Jan 2024 – Jul 2025*
Tech Stack: Python, React, MongoDB, Docker, FFmpeg, AWS

- Reduced client onboarding time for CCTV integrations by **40%** by building a scalable multi-protocol video ingestion pipeline supporting WebRTC, RTSP and HLS. Strengthened SDLC practices through code reviews and technical documentation
- Optimized the frame processing pipeline by integrating HLS-based streaming to deliver processed video output **3x** faster to users
- Slashed video preprocessing latency from **180s** to **50s** across **~150GB/day** of video data by revamping the FFmpeg pipeline into a maintainable, multithreaded, I/O-tuned architecture with asynchronous S3 prefetching for real-time WebRTC streams
- Improved object detection accuracy by **20%** by building a full-stack, human-in-the-loop annotation platform with React and FastAPI for reviewing and correcting model predictions

Infibeam Avenues — **Bangalore, India**
Software Development Engineer — *Jul 2022 – Jan 2024*
Tech Stack: Ruby on Rails, AngularJS, MySQL, Docker, ELK, Kubernetes

- Shipped a Firebase-based seller notification system serving **1M+** sellers, driving a **15%** increase in seller engagement through real-time push notifications within a cross-functional Agile team
- Saved support teams **120+ hours/month** by spearheading a suite of REST APIs in Ruby on Rails to automate seller ticketing workflows, collaborating with third-party partners and integrating external services
- Enabled admins to process **2K+** seller stock approval requests per day by delivering a pull-based work allocation system. Distributed requests across admins and documented allocation logic, request states, and edge cases for support teams

PROJECTS

Distributed Log Processing System | Next.js, FastAPI, ClickHouse, GCP [Code]

- Accelerated end-to-end log analysis throughput by **10x** to **~1M events/min** compared to a synchronous baseline by engineering an Apache Kafka + Flink ETL pipeline and an observability dashboard with <100ms query latency via Redis caching

HyRA - Hybrid Routing Agent | React, Flask, MySQL, GCP [Code]

- Developed a cost-aware LLM routing service with vLLM and LangGraph, improving routing accuracy from **71%** to **89%** on a curated evaluation set through an LLM-as-a-Judge feedback loop and iterative prompt tuning
- Productionized model serving on GCP with Langfuse-based monitoring and performance profiling for distributed LLM workloads

SKILLS

Languages & Databases: Python, Go, Ruby, Java, TypeScript/JavaScript, MySQL, PostgreSQL, MongoDB, Redis
Web & App Development: HTML/CSS, React, Angular, Ruby on Rails, Flask, FastAPI, gRPC
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Terraform, Nginx, ELK Stack, Prometheus, Grafana, Linux/Unix
LLM & Developer Tools: OpenAI API, Claude, Copilot, Visual Studio, Cursor, Git/GitHub, Postman